

Artificial Neural Networks

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Historical background

- ▶ 1943 - McCulloch and Pitts, artificial neuron model
- ▶ 1949 - Hebb, biological neural networks can remember information
- ▶ 1958 - Rosenblatt, perceptron model
- ▶ 1960 - Widrow and Hoff, first microcomputer („Madeline”)

Historical background

- ▶ 1969 - Minsky and Papert, generalised the limitations of single layer Perceptrons to multilayered systems (XOR problem)
- ▶ 1986 - McClelland, Rumelhart, back propagation algorithm
- ▶ 1990s - great progress

Biological-Type Neural Networks

Biological neuron

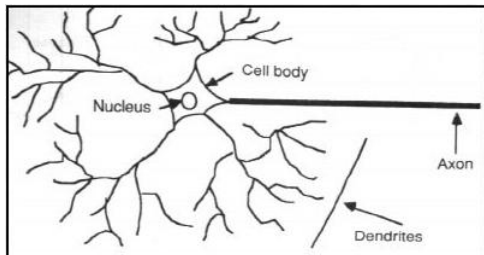


Figure: Components of biological neuron.

Biological-Type Neural Networks

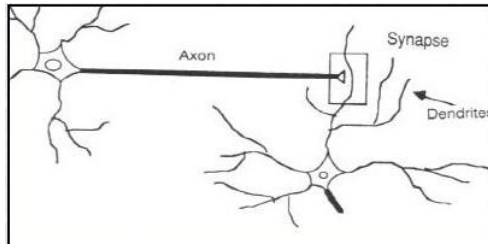


Figure: The synapse.

Biological-Type Neural Networks

- ▶ Neuron collects signals from others through a host of fine structures called dendrites.
- ▶ The neuron sends out spikes of electrical activity through a long, thin stand known as an axon, which splits into thousands of branches.
- ▶ At the end of each branch, a structure called a synapse converts the activity from the axon into electrical effects that inhibit or excite activity from the axon into electrical effects that inhibit or excite activity in the connected neurones.
- ▶ When a neuron receives excitatory input that is sufficiently large compared with its inhibitory input, it sends a spike of electrical activity down its axon.
- ▶ Learning occurs by changing the effectiveness of the synapses so that the influence of one neuron on another changes.

From Human Neurones to Artificial Neurones

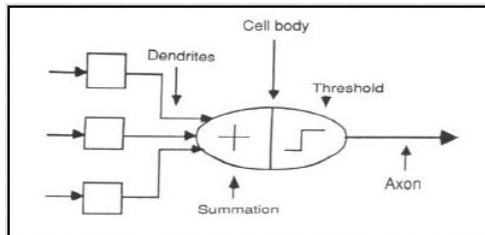


Figure: The neuron model

A simple neuron

- ▶ An artificial neuron is a device with many inputs and one output.
- ▶ The neuron has two modes of operation: the training mode and the using mode.
- ▶ In the training mode, the neuron can be trained to fire (or not), for particular input patterns.
- ▶ In the using mode, when a taught input pattern is detected at the input, its associated output becomes the current output.
- ▶ If the input pattern does not belong in the taught list of input patterns, the firing rule is used to determine whether to fire or not.

A simple neuron

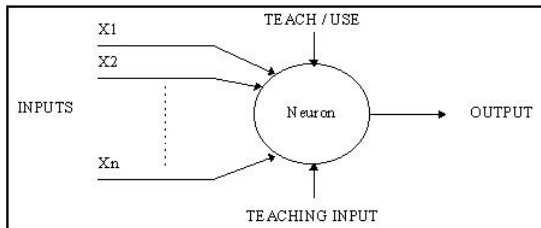


Figure: The neuron model

A more complicated neuron

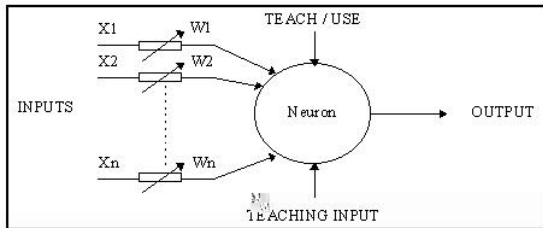


Figure: McCulloch and Pitts model (MCP)

An artificial neuron

A Single Neuron

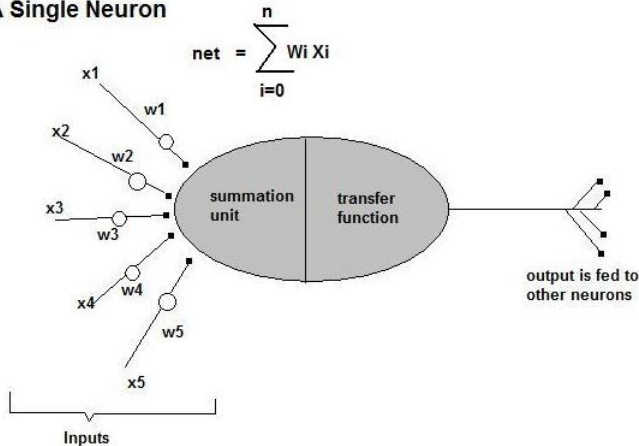


Figure: McCulloch and Pitts model (MCP)

Basis Function and Activation Function

For an analytical study, the connection networks are mathematically represented by a basis function $u(w, x)$, where w stands for the *weight matrix*, and x for the *input vector*. The basis function has two common forms:

- ▶ Linear-Basis function (LBF) is a hyperplane-type function. This is a first-order basis function. The net value is a linear combination of the inputs
- ▶ Radial-basis function (RBF) is a hypersphere-type function. This involves a second-order (nonlinear) basis function. The net value represents the distance to a reference pattern

$$u_i(w, x) = \sum_{j=1}^n w_{ij} x_j$$
$$u_i(w, x) = \sqrt{\sum_{j=1}^n (x_j - w_{ij})^2}$$

Activation Function (Neuron Function/Transfer function)

- ▶ The net value as expressed by the basis function, $u(w, x)$, will be transformed by a nonlinear activation function of the neuron.
- ▶ Often, the activation function is a nondecreasing function, although activation functions are not restricted to nondecreasing functions.
- ▶ Generally, some sort of threshold function is used: a hard limiting threshold function (a sgn function), or a linear or semi-linear function, or a smoothly limiting threshold
- ▶ For this smoothly limiting function often a sigmoid (S-shaped) function

Activation Function

The choice of the activation function depends on the problem neural network is designed to solve.

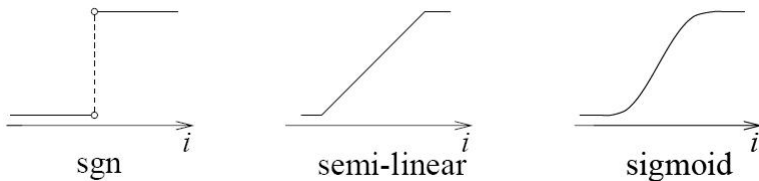


Figure: Different types of activation function

Activation Function

- ▶ For linear units, the output activity is proportional to the total weighted output.
- ▶ For threshold units, the output is set at one of two levels, depending on whether the total input is greater than or less than some threshold value.
- ▶ For sigmoid units, the output varies continuously but not linearly as the input changes. Sigmoid units bear a greater resemblance to real neurones than do linear or threshold units, but all three must be considered rough approximations.

Activation Function

<i>Name</i>	<i>Function</i>	<i>Derivative</i>
Sigmoid	$f(x) = \frac{1}{1+e^{-\beta x}}$	$f'(x) = \beta(1 - f(x))f(x)$
Hyperbolical tangent	$f(x) = \tanh(\beta x)$	$f'(x) = \beta(1 - f^2(x))$
Sinus	$f(x) = \sin(\beta x)$	$f'(x) = \beta\sqrt{1 - f^2(x)}$
Cosinus	$f(x) = \cos(\beta x)$	$f'(x) = -\beta\sqrt{1 - f^2(x)}$
$\frac{x}{1+ x }$	$f(x) = \frac{\beta x}{1+ \beta x }$	$f'(x) = \frac{\beta}{1+ \beta x } - \frac{\beta x}{(1+ \beta x)^2}$

Nonlinear activation functions and their derivatives

Structures of Neural Networks

- ▶ The most common type of artificial neural network consists of three groups (layers) of units:
 - ▶ a layer of "input" units is connected to
 - ▶ a layer of "hidden" units, which is connected to
 - ▶ a layer of "output" units
- ▶ The activity of the input units represents the raw information that is fed into the network.
- ▶ The activity of each hidden unit is determined by the activities of the input units and the weights on the connections between the input and the hidden units.
- ▶ The behaviour of the output units depends on the activity of the hidden units and the weights between the hidden and output unit

Structure of Neural Networks

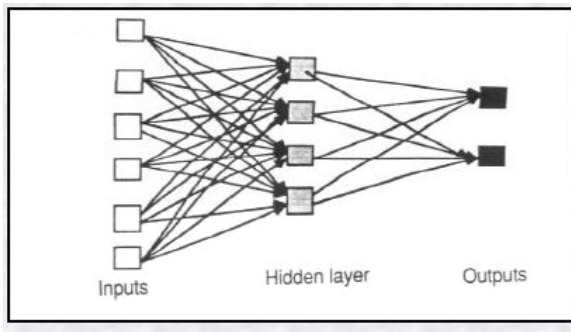


Figure: An example of a simple feed forward network

- ▶ **Supervised** learning which incorporates an external teacher, so that each output unit is told what its desired response to input signals ought to be.
- ▶ **Unsupervised** learning uses no external teacher and is based upon only local information.
It is also referred to as self-organisation, in the sense that it self-organises data presented to the network and detects their emergent collective properties.

Learning methods

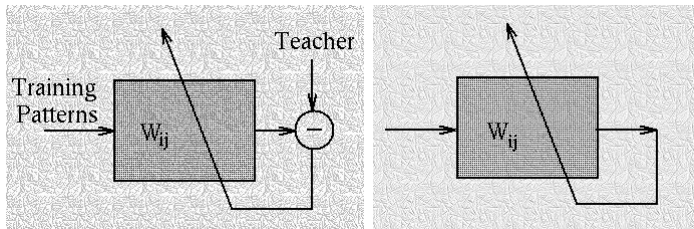


Figure: Supervised and unsupervised learning

Backpropagation

- ▶ Supervised learning (with teacher)
- ▶ Two data sets: learning and testing
- ▶ Learning set : learning vector
Input vector
Output vector
- ▶ Obtained output is compared with the desire output
- ▶ If the answer is wrong, the error of the answer is counted
- ▶ The error is passed backward
- ▶ New weights are counted

Applications fields

- ▶ association
- ▶ clustering
- ▶ classifications
- ▶ pattern completion
- ▶ regression and generalization
- ▶ optimization

Applications

- ▶ Pattern recognition
- ▶ Classification of the objects
- ▶ Sales forecasting, risk management
- ▶ Industrial process control
- ▶ Customer research
- ▶ Data validation
- ▶ Medical Modelling and Diagnosing
- ▶ Approximation of functions

Conclusions

- ▶ Wide range of applications
- ▶ Applied there, where traditional (algorithmic) methods are not sufficient
- ▶ Artificial neural network with three hidden layers can solve almost all problems